

# THEJUS MAHAJAN PH. D.

Bioinformatics & Data Scientist



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## CAREER INTERESTS

Computational Scientist (PhD) with 5+ years of experience in Biological Systems Modeling and High-Dimensional Data Analysis. Experienced during post-doc in applying stochastic population models in hindcast and forecast projections. Trained in Python and R pipelines for noisy data analysis. Seeking for a career shift to Bioinformatics. I am leveraging on my recent Bioinformatics and Biostatistics training along with the advanced mathematical modeling expertise gained during the Ph. D. and post-doc period to solve problems in human metabolic networks and clinical data analysis.

## EXPERIENCE

### Further Training – Bioinformatics

08/2025 – Present

CQ Beratung + Bildung, Berlin

- Introduction to programming and Python, including Shell (BASH), programming methodology, and object-oriented programming in Python.
- Bioinformatics and databases, including NCBI, Biopython, web services, and database management with SQL.
- Applied bioinformatics, including sequence analysis and strategies, structural bioinformatics, software for structure visualization, and NGS analysis using Nextflow and Galaxy.
- Applied biostatistics, including statistical analysis using R, descriptive and multivariate statistics, ANOVA, PCA, HCA, workflows using R/Bioconductor, and Markov chain implementation.

### Internship – Machine Learning in R

02/2026 – 04/2026

HealthTwist GmbH, Berlin

- Implementation of Machine Learning workflows using the Tidymodels framework in R.
- Supervised by Dr. Andreas Busjahn.
- Practical application of bioinformatics and biostatistical analysis skills.

### Guest Scientist

05/2025 – 10/2025

Helmholtz-Zentrum Hereon, Geesthacht

### Computational Biologist & Systems Modeler

- Spearheaded a complex research project modeling evolutionary processes in marine environments, in collaboration with the Ecosystem Modeling Department.
- Applied advanced computational techniques to simulate the long-term impacts of environmental changes on marine ecosystems.

### Post-doctoral Scientist

08/2021 – 01/2025

University of Hamburg, Hamburg

- Model development: Developed the novel “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework to project future climate warming scenarios.

## STRENGTHS

### Advanced Scientific Modelling

Developed and implemented a novel biogeochemical model (CLC) within the GOTM-FABM framework.

### Complex Problem-Solving

Resolved a critical data-loss error by creating a Fortran simulation to correct for a misplaced particle detector.

### Bioinformatics

Successfully bridged physical science expertise with modern biological workflows.

## SKILLS

|                |           |
|----------------|-----------|
| Python         | ● ● ● ● ● |
| R (Tidymodels) | ● ● ● ● ● |
| Nextflow       | ● ● ● ● ● |
| SQL            | ● ● ● ● ● |
| Fortran        | ● ● ● ● ● |
| LaTeX          | ● ● ● ● ● |

## LANGUAGES

|             |           |
|-------------|-----------|
| English     | ● ● ● ● ● |
| German (B1) | ● ● ● ● ● |
| French      | ● ● ● ● ● |
| Malayalam   | ● ● ● ● ● |
| Tamil       | ● ● ● ● ● |

## REFEREES

### Dr. Andreas Busjahn

HealthTwist GmbH

andreas@busjahn.net

### Prof. Dr. Kai Wirtz

Helmholtz-Zentrum Hereon

kai.wirtz@hereon.de

Hamburg, den 05.02.2026

- High-performance computing: Engineered models in Fortran and analyzed large NetCDF datasets using Python in a Linux environment.
- Statistical analysis and visualization of ecological time-series data using R.

**Ph.D. Researcher Astrochemistry**

📅 10/2015 – 09/2018

*Univ. Paris-Saclay, France*

- Big Data Analysis: Acquired and processed terabytes of raw data from tandem particle accelerator experiments.
- Algorithm Development: Utilized C++ for noise cancellation, signal smoothing.
- Simulation: Developed a molecular fragmentation model for the KIDA database.

## EDUCATION

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**Ph.D. in Astrochemistry**

2015 – 2018

Université Paris-Saclay, France

**M.Sc. in Physics**

2012 – 2015

National Institute of Technology Calicut, India

CGPA: 8.2/10

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Thejus Mahajan